Text-to-Speech Synthesis Using Linear Predictive Coding

Speech Processing

CMPE 532

Homework 2

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1 Introduction and Motivation

Text-to-speech (TTS) is a technology that automatically turns written text into spoken words. It is one of the oldest and most useful areas of digital speech study. TTS is used for many important tasks, such as helping people with vision loss read text or providing voices for phones and smart devices. The goal of any TTS system is to sound clear and natural, with the correct rhythm, emphasis, and tone.

Among the many ways to create speech, Linear Predictive Coding (LPC) is one of the most important historical methods. This method is based on the "source-filter" model, which mimics how the human body produces sound. LPC represents the human vocal tract using a small set of mathematical numbers. Because LPC describes speech with these few numbers, the system can build speech from a model rather than needing to store large recordings of human voices.

This report explains a basic LPC-based TTS system. This system follows the classic methods described by Rabiner and Schafer in their textbook [7]. The goal is to understand how the different parts of LPC work together to create speech, how the math works at each step, and what the limits of this approach are.

2 System Architecture and Components

A basic LPC-based TTS system can be decomposed into two broad stages: a front-end that processes text into a sequence of phonetic and prosodic targets, and a back-end that uses LPC synthesis to generate the actual speech waveform. Figure 1 illustrates the high-level pipeline.

```
      [Input Text]
           |
           v
+---------------------+
|   Text Analysis     |  <- Tokenization, normalization
|   (Front-End)       |
+----------+----------+
           |
           v
+---------------------+
|  Phonetic Transcr.  |  <- Grapheme-to-phoneme (G2P)
|  & Prosody Modeling |  <- Duration, pitch (F0) rules
+----------+----------+
           |  Phoneme sequence + F0 + duration targets
           v
+---------------------+
|   LPC Parameter     |  <- Per-phoneme LPC coefficients
|   Lookup / Interp.  |     from pre-analyzed inventory
+----------+----------+
           |
           v
+---------------------+
|  Excitation Signal  |  <- Voiced: periodic pulse train
|  Generation         |  <- Unvoiced: white noise
+----------+----------+
           |
           v
+---------------------+
|  LPC Synthesis      |  <- All-pole filter driven by
|  Filter (All-Pole)  |     excitation; outputs speech
+----------+----------+
           |
           v
     [Output Speech Waveform]
```

Figure 1: High-level LPC-based TTS pipeline.

2.1 Text Analysis and Normalization

The front-end begins with raw text normalization: expanding abbreviations, numbers, and punctuation into their spoken forms. This is a purely linguistic step with no signal processing content, but it is critical for correctness.

2.2 Grapheme-to-Phoneme Conversion (G2P)

The normalized text is converted into a sequence of phonemes using a G2P module — either a rule-based lookup table or a pronunciation dictionary. For a basic English system, a pre-compiled lexicon (such as the CMU Pronouncing Dictionary) is sufficient. The

output is a sequence of phoneme labels along with coarse syllable and word boundary markers.

In the code for this assignment, G2P is implemented using NLTK's CMU dictionary and a simple rule-based fallback for out-of-vocabulary words. The resulting phoneme sequence serves as the input to the LPC parameter lookup.

2.3 Prosody Modeling

A simple rule-based prosody module assigns:

- **Duration** to each phoneme (e.g., stressed vowels are longer than unstressed ones),
- **Fundamental frequency (F0)** contour over the utterance (e.g., a declination rule that starts F0 high and drops toward the end of sentences).

In a basic system, these are implemented as lookup tables with simple interpolation between phoneme boundaries.

Rules are implemented to code by adding a 20 ms bonus to vowel durations and linearly decreasing the pitch from 1.10 to 0.88 times the base frequency over the course of the utterance.

2.4 LPC Parameter Store

The core of the back-end is a pre-built inventory of LPC parameter sets, one per phoneme (or a small set per phoneme to capture allophonic variation). Each entry stores:

- The LPC predictor coefficients $\mathbf{a} = [a_1, a_2, \ldots, a_p]$ of order p (typically $p = 10$–12 for telephone-bandwidth speech),
- The residual gain G, which controls output amplitude,
- A voiced/unvoiced (V/UV) flag.

In the implementation, these parameters are synthesized from formant frequency data based on [5, 2].

2.5 Excitation Generation and LPC Synthesis Filter

At runtime, for each synthesis frame:

1. If voiced, a periodic impulse train at the target F0 period is generated.
2. If unvoiced, white Gaussian noise is generated.
3. The excitation is scaled by G and passed through the all-pole synthesis filter.

Smooth transitions between consecutive phonemes are achieved by linearly interpolating the LPC coefficients frame-by-frame across phoneme boundaries, preventing discontinuous spectral jumps.

3 Signal Processing Fundamentals

3.1 The Source-Filter Model

LPC-based synthesis is grounded in the source-filter model of speech production [7, 4]. In this model, the speech signal $s(n)$ is viewed as the output of a linear, time-varying filter $H(z)$ (representing the vocal tract) driven by a source excitation $e(n)$:

$$S(z) = H(z) \cdot E(z). \tag{1}$$

For voiced sounds, $e(n)$ is a quasi-periodic glottal pulse train; for unvoiced sounds, it approximates white noise. The vocal tract filter $H(z)$ is modeled as an all-pole system:

$$H(z) = \frac{G}{1 - \sum_{k=1}^{p} a_k z^{-k}} = \frac{G}{A(z)}, \tag{2}$$

where $A(z)$ is the LPC prediction polynomial of order p.

3.2 LPC Analysis

Given a short (typically 20–30 ms) windowed frame of speech, the predictor coefficients are obtained by minimizing the mean squared prediction error:

$$\hat{s}(n) = \sum_{k=1}^{p} a_k \, s(n-k). \tag{3}$$

The error $e(n) = s(n) - \hat{s}(n)$ is the LPC residual, which approximates the source excitation. The optimal coefficients satisfy the Yule-Walker equations:

$$\mathbf{R}\,\mathbf{a} = \mathbf{r}, \tag{4}$$

where $\mathbf{R}$ is the $p \times p$ autocorrelation matrix of the speech frame and $\mathbf{r}$ is the vector of autocorrelation lags $r(1)$ through $r(p)$. The Levinson-Durbin recursion solves this system efficiently in $\mathcal{O}(p^2)$ operations, making LPC analysis computationally practical [1, 7].

3.3 Spectral Envelope Representation

The magnitude spectrum of the all-pole filter $|H(e^{j\omega})|$ approximates the spectral envelope of the original speech frame. The LPC order p determines the resolution of this envelope: a higher order captures more spectral detail (more formants), while a lower order gives a coarser smoothing. For speech with a sampling rate of 8 kHz (telephone quality), $p = 10$ is generally sufficient to represent the first four or five formant resonances.

3.4 Windowing and Frame Overlap

Because the vocal tract changes continuously, LPC parameters must be updated over time. In practice, analysis frames are extracted every 10 ms with a 25–30 ms window (Hamming or Hanning), producing a new parameter set at each frame boundary. During synthesis, parameters are interpolated linearly between consecutive frames to ensure smooth temporal transitions.

3.5 Voiced/Unvoiced Decision and Pitch Estimation

Correctly classifying each frame as voiced or unvoiced is essential for naturalness. A basic approach uses the normalized autocorrelation of the LPC residual: a strong peak at a lag τ indicates periodicity with fundamental period $T_0 = \tau / f_s$. In synthesis, the target F0 (as specified by the prosody module) directly sets the period of the impulse train excitation.

4 Evaluation and Limitations

4.1 Intelligibility

Basic LPC TTS systems achieve acceptable intelligibility for simple, clearly structured utterances. Listeners can generally understand the output, particularly in constrained vocabularies or predictable sentence contexts. However, intelligibility degrades for longer utterances, polysyllabic words with complex consonant clusters, and language-specific phonotactic structures — especially for morphologically rich languages such as Turkish, where vowel harmony and agglutination produce phoneme sequences that may not be well-covered by a small phoneme inventory.

4.2 Naturalness and Voice Quality

The most prominent weakness of basic LPC synthesis is the characteristic "robotic" or "buzzy" quality of the output. This arises from several sources:

- **Excitation oversimplification:** Modeling the glottal source as a pure impulse train ignores the rich spectral shape of the real glottal pulse. More realistic excitation models (e.g., the Liljencrants-Fant glottal model) substantially improve perceived naturalness but add complexity.
- **Coarticulation neglect:** The system assigns one parameter set per phoneme in isolation. In real speech, phonemes overlap temporally — a vowel is already being shaped while the preceding consonant is being articulated. Simple coefficient interpolation at boundaries is a crude approximation of this process.
- **Prosody flatness:** A rule-based F0 contour is far from the rich intonation of natural speech. Monotone or mechanically declining F0 is immediately recognizable as synthetic.

4.3 LPC Order and Spectral Fidelity

Selecting too low an order causes over-smoothing, merging distinct formants and producing a muffled timbre. Too high an order introduces spurious spectral peaks and can model microstructure (fine pitch harmonics) rather than the vocal tract envelope, degrading synthesis quality. In practice, the optimal order must be tuned empirically for each language and speaker.

4.4 Comparison with Other Basic Methods

Diphone concatenation, another classical approach, produces more natural-sounding output because it stores real speech transitions between phoneme pairs, preserving natural

coarticulation. However, it requires substantially larger storage and a full recorded diphone inventory. LPC synthesis, by contrast, has minimal storage requirements — a phoneme inventory of 50 entries each storing 12 coefficients is on the order of kilobytes — making it well suited for resource-constrained embedded systems.

4.5 Quantitative Evaluation

Formal evaluation of TTS systems employs [3, 6]:

- **Mean Opinion Score (MOS):** Listeners rate naturalness on a 1–5 scale. Basic LPC systems typically score in the 2.5–3.5 range — intelligible, but clearly synthetic.
- **Word Error Rate (WER) via ASR:** Intelligibility can be measured by passing synthesized speech through an automatic speech recognizer and measuring recognition error.
- **Formant tracking accuracy:** For diagnostic purposes, the F1/F2 trajectories of synthesized vowels can be compared against natural speech via spectrographic analysis.

5 Conclusion

LPC-based text-to-speech synthesis represents a foundational, analytically transparent approach to speech generation. By applying the source-filter model — the same model that underlies much of classical speech analysis — it provides a direct bridge between the signal processing theory developed in Rabiner and Schafer [7] and a working synthesis application. The system converts text to phoneme sequences via a front-end, retrieves pre-analyzed LPC parameters for each phoneme, drives an all-pole filter with an appropriate excitation signal, and produces an intelligible speech waveform with modest computational and storage requirements.

The limitations of this basic approach — robotic voice quality, coarticulation neglect, and simplistic prosody — are precisely the problems that motivated the evolution toward more sophisticated methods: mixed excitation LPC, formant synthesis with detailed glottal models, diphone and unit selection concatenation, and eventually the statistical parametric (HMM/DNN-based) and end-to-end neural synthesis systems that dominate modern TTS. Understanding LPC synthesis in its basic form thus provides the essential vocabulary and conceptual scaffolding for appreciating why and how each of these advances was necessary.

References

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